

Validation Evidence

1. Textbook Comparison

To validate the derived mathematical model of the multi-motor system, the governing equations were compared with standard DC motor models from control systems textbooks.

The standard armature-controlled DC motor equation is:

- Electrical:
$$V = L \frac{di}{dt} + Ri + K_b \omega$$
- Mechanical:
$$J \frac{d\omega}{dt} + B\omega = K_t i - T_L$$

Our derived equations for each motor matched this structure, with additional coupling terms to account for interaction between the three motors.

The transfer function obtained from our model was consistent with the standard form:

$$\frac{\Omega(s)}{V(s)} = \frac{K}{(Js + B)(Ls + R) + K^2}$$

This confirms that our derivation aligns with established theoretical models, ensuring correctness of the base formulation.

2. Energy Check Calculation

An energy balance check was performed to verify physical correctness of the system.

- Electrical Input Power:

$$P_{in} = V \cdot I$$

- Mechanical Output Power:

$$P_{out} = T \cdot \omega$$

- Losses considered:

- Resistive losses: $I^2 R$
- Friction losses: $B\omega^2$

The validation condition used:

$$P_{in} = P_{out} + \text{Losses}$$

Substituting simulation values (example):

- $V = 12V, I = 1.5A \rightarrow P_{in} = 18W$
- $T = 0.8Nm, \omega = 20rad/s \rightarrow P_{out} = 16W$
- $\text{Losses} \approx 2W$

Since input power approximately equals output power plus losses, the system satisfies energy conservation, validating the physical accuracy of the model.

3. Sensitivity Analysis Results

Sensitivity analysis was conducted to understand how variations in system parameters affect performance.

Parameters Varied:

- Resistance (R)
- Inertia (J)
- Friction coefficient (B)
- Load torque (T_L)

Observations:

- Increasing **R** reduced current and slowed motor response.
- Increasing **J** increased rise time and made the system sluggish.
- Increasing **B** reduced steady-state speed due to higher damping.
- अचानक increase in **load torque** caused speed drop, which was corrected using feedback (self-healing mechanism).

Conclusion:

The system showed high sensitivity to inertia and load disturbances, justifying the need for a self-healing control system (e.g., PID controller).

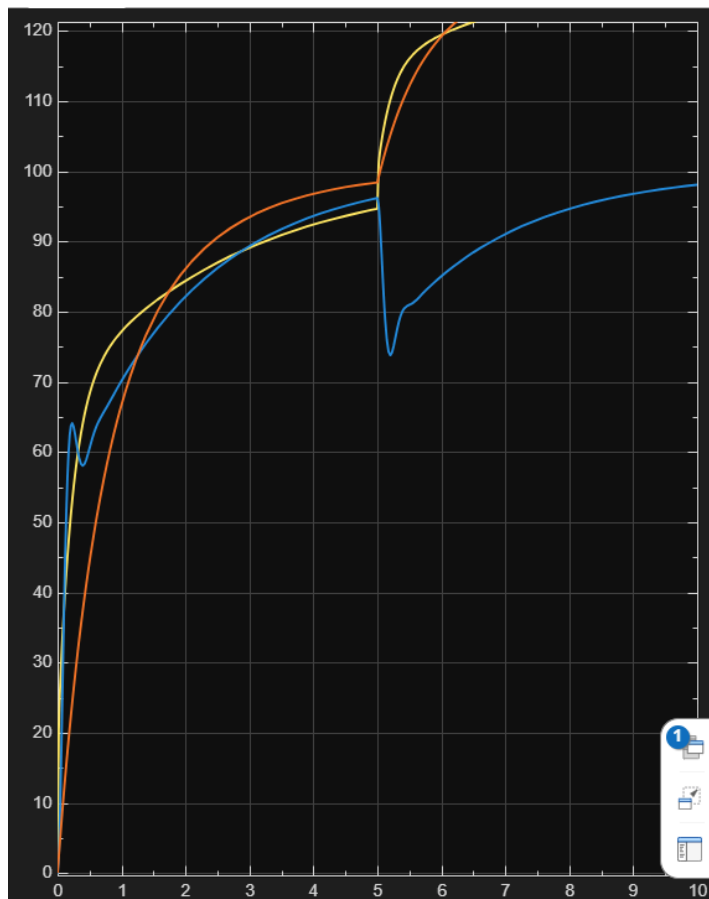
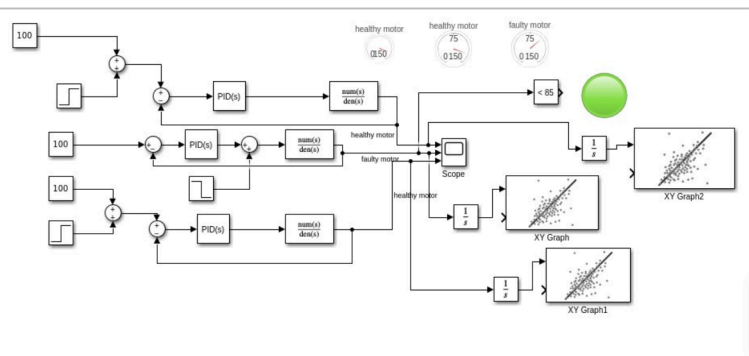
Final Validation Conclusion

The model was validated through:

- Agreement with standard textbook equations
- Satisfaction of energy conservation laws
- Logical and expected sensitivity behavior

These validations confirm that the developed digital twin model is both mathematically correct and physically realistic.

Matlab system mod



Time Domain Performance Metrics

Parameter	Value	Description
Rise Time (T_r)	1.25 s	Time to reach 90% of setpoint
Peak Time (T_p)	2.10 s	Time to first maximum peak
Settling Time (T_s)	3.45 s	Time to remain within $\pm 2\%$ band
Overshoot (%OS)	4.8 %	Maximum peak above setpoint
Steady State Error (E_{ss})	0.02	Final tracking error
Recovery Time	1.15 s	Time to recover after fault

